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# When personalization deters delegation to algorithms:

Evidence from a **preregistered** experiment

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— 23<sup>rd</sup> June, 2026, Symposium on ‘AI and the Future of Work’, Berlin —

Preregistration: <https://aspredicted.org/7un5qf.pdf>

- ▶ Huge & unconnected literature on algorithm/automation/AI reliance (→ review by Chugunova & Sele, 2022)
  - ▶ Common assumption: **personalisation** (tailoring to the user's preferences) should **increase** acceptance of an algorithmic decision aid
  - ▶ But personalisation may also be **double-edged**: a system that “decides as you would” substitutes for personal judgment more directly than an impersonal rule
- Does that backfire? Does personalisation actually raise *behavioural* willingness to delegate?

## Well-known (Swiss) robo-advisors(' marketing strategies)

- ▶ “We rely on a low-cost, passive investment strategy and have selected a total of 9 equity, bond and real estate ETFs from over 1,500 ETFs (exchange-traded funds) and put them together in such a way that they **achieve an optimal risk/return ratio.**” (findependent.ch; for GER, see e.g., whitebox.eu)
- ▶ “Chat with Selma: Selma **looks at your financial life and learns how you should structure your finances.** Get your plan: Selma **creates a personalized investment plan that fits your life.** Sit back and relax! Selma takes care of your money from now on and does all the buying and selling for you.” (selma.com; for GER, see e.g., quirion.de)



- ▶ Does algorithm reliance differ for **objective optimization vs. preference-based personalisation** in a preferential-choice environment?
  - (i) ...at first contact?
  - (ii) ...overall?

- ▶ What do we mean by **objective optimization**?
  - an algorithm that chooses some ‘objectively ideal’ option independently of a subject’s preferences (here: **highest EV**)
- ▶ What do we mean by **personalised**?
  - an algorithm that measures a subject’s preferences and chooses accordingly (here: **implementation of revealed preferences**)
- ▶ “Objective” is a *label*, not a normative claim: we are not asserting that EV-maximisation is rational, nor that revealed preferences are perfectly measured.



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  - an algorithm that measures a subject’s preferences and chooses accordingly (here: **implementation of revealed preferences**)
- ▶ In the experiment, we describe both algorithms in intuitive terms:

**EV:** *‘The algorithm works out how much each option would yield on average and selects the option for which this is highest.’*

**Prefs:** *‘The algorithm analyses your choices from Part 1 and selects the best option for you in light of your Part-1 choices.’*

## Part 1 – Elicitation of preferences

- ordered lotteries, no foils, unlimited time
- “hypothetical, but the choices may be meaningful for Part 2”

## Part 2 – Lottery Selection Task

- could stand for an investment task
- complexity (number of options, outcomes under different SOTW)
- time pressure (think: relative to the number of options)
- each Part-2 task built from 2 **anchor options** (transformed from Part 1) + 6 **decoys** (each dominated by an anchor, but not obviously so)

|          | State 1 | State 2 | State 3 | State 4 | State 5 | State 6 | State 7 | State 8 | State 9 | State 10 |
|----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
| Option 1 | 225     | 225     | 225     | 225     | 225     | 225     | 225     | 205     | 0       | 0        |
| Option 2 | 525     | 525     | 525     | 525     | 525     | 0       | 0       | 0       | 0       | 0        |

| State:   | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Option 1 | 225 | 225 | 0   | 225 | 0   | 225 | 225 | 225 | 225 | 0   |
| Option 2 | 525 | 0   | 505 | 0   | 525 | 0   | 0   | 0   | 525 | 505 |
| Option 3 | 525 | 0   | 525 | 0   | 525 | 0   | 0   | 0   | 525 | 525 |
| Option 4 | 225 | 225 | 0   | 225 | 205 | 205 | 205 | 225 | 225 | 0   |
| Option 5 | 225 | 205 | 0   | 205 | 205 | 225 | 225 | 205 | 225 | 0   |
| Option 6 | 225 | 225 | 0   | 225 | 205 | 225 | 225 | 225 | 225 | 0   |
| Option 7 | 525 | 0   | 525 | 0   | 525 | 0   | 0   | 0   | 225 | 525 |
| Option 8 | 505 | 0   | 525 | 0   | 505 | 0   | 0   | 0   | 505 | 525 |

- ▶ 8 incentivised rounds; 4 filler tasks interleaved in Part 1
- ▶ Per round: 5 s pre-view → 5 s delegate/not → up to 8 s own decision



- ▶ **Recruitment:** Prolific UK, online (z-Tree unleashed); preregistered target 120/treatment before exclusions.
- ▶ **Exclusions (preregistered):** bot/slider checks, comprehension after 2nd attempt, self-reported AI use, implausibly fast log-times, incomplete data.
- ▶ **Final sample:** **N = 233** (EV:  $n = 118$ , Prefs:  $n = 115$ ).
- ▶ **Incentives:** GBP 2.00 base + one randomly drawn Part-2 round (1 penny / point); up to GBP 8.90 bonus. Mean earning GBP 4.14, median duration ~16 min.
- ▶ **Ethics:** IRB approval by GfEW (cert. 19HBJNwQ); informed consent.



- ▶ **H1.** People are more likely to delegate financial decisions to an **EV-maximising** algorithm than to a **personalised** algorithm that incorporates their (revealed) preferences.
- ▶ **Primary tests (preregistered):**
  - (1) One-sided **Boschloo** exact test on **Round-1** delegation (first contact)
  - (2) One-sided **Wilcoxon–Mann–Whitney** on participant-level mean delegation across Part 2
- ▶ **Preregistered LPM:** decision-set FE, participant-clustered SE, treatment  $\times$  period interaction, share of prior delegated rounds with zero payoff.
- ▶ **Exploratory:** categorical *never-delegation*, win-stay/lose-shift, ex-post reason items.
- ▶ Preregistration: <https://aspredicted.org/7un5qf.pdf>

# Main Results: First Contact

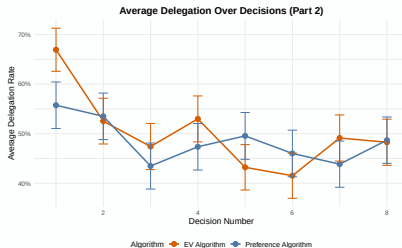


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- ▶ **Benchmark:** only ~34–35% of *self-chosen* options were non-dominated; delegated rounds paid on average ~25% more than self-decided ones  $\Rightarrow$  delegation is the instrumentally rational thing to do.
- ▶ **Round 1: 66.9% (EV) vs. 55.8% (Prefs).**
- ▶ Boschloo exact test, one-sided:  
 $p = 0.0413 \Rightarrow$  **H1 supported at first contact.**
- ▶ Treatment gap concentrated in the first decision.



# Main Results: Repeated Decisions

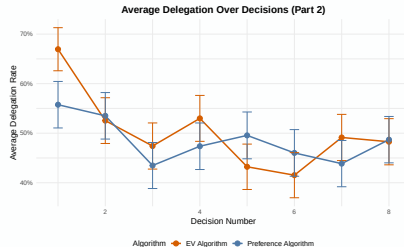


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- ▶ **Average delegation:** no significant treatment difference overall (WMW one-sided  $p = 0.2425$ ).
- ▶ Preregistered LPM (decision-set FE, clustered SE):
  - EV main effect:  $\beta = -0.049$ ,  $p = 0.56$
  - EV  $\times$  period:  $\beta = 0.002$ ,  $p = 0.87$
  - Share of prior zero-payoff delegated rounds:  
 $\beta = -0.385$ ,  $p < 0.001$
- ▶ Convergence after Round 1 driven **more by a decline in EV** than by a rise in Prefs.
- ▶ But: first-contact decisions can determine whether users *come back at all*.



# Categorical Non-Delegation (not preregistered)

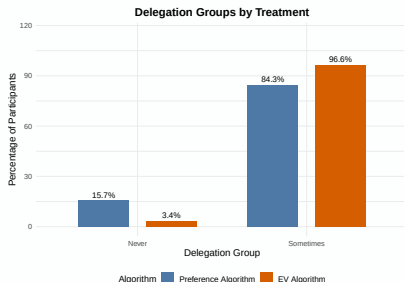


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- ▶ Never-delegators:
  - **15.7%** in Prefs (18/115)
  - **3.4%** in EV (4/118)
- ▶ Always-delegators show the opposite, weaker pattern (13.6% Prefs vs. 19.1% EV).
- ▶ Never-delegators do *not* differ in age, income, education, gender, time spent, financial risk-taking, AI usage, or reliability.
- ▶ Within Prefs: never-delegators report (weakly) *higher* satisfaction with Part-1 choices ( $\Rightarrow$  not driven by dislike of the elicited profile).
- ▶ **Replication** in earlier (PhD-thesis) study: 32/189 vs. 17/187, Boschloo  $p = 0.0135$ .



# Mechanisms: *What deters delegation differently by algorithm*

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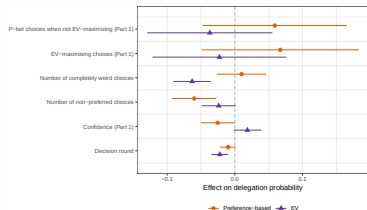
- ▶ LPM of individual delegation on *expected non-preferred* and *expected “completely weird”* choices,  $\times$  treatment, controlling for round, Part-1 confidence, risk preferences, task FE:

- ▶ Expected **non-preferred** choices:
  - Prefs:  $\beta = -0.058$ ,  $p < 0.001$
  - EV interaction:  $+0.035$ ,  $p = 0.09$

- ▶ Expected **“completely weird”** choices:

- Prefs:  $\beta = +0.007$ , ns
- EV interaction:  $-0.070$ ,  
 $p = 0.003$

- ▶ *Non-preferred* deters under **Prefs**;  
*extreme errors* deter under **EV**.



95% confidence intervals. Linear probability model with task fixed effects and part EV-condition effects are main effect + interaction term.

- ▶ Differences run more by *delegator type* than by treatment.
- ▶ **Delegators** (vs. never-delegators) more often cite performance motives (70.1% vs. 36.4%: “computer would choose better”).
- ▶ **Never-delegators** more often cite distrust of the algorithm and responsibility for the choice (63.6% vs. 38.9%).
- ▶ Crucially: they want responsibility for the **choice**, *not* for the **outcome** ⇒ this is a positive valuation of **choice authority**, not blame-avoidance.
- ▶ Items on being algorithmically analysed and on opacity (“don’t know how it chooses”) are rarely selected ⇒ not a transparency story.
- ▶ Consistent with *control / decision-autonomy* as a driver of acceptance (Schaap et al., 2024; Ivanova-Stenzel & Tolksdorf, 2025).

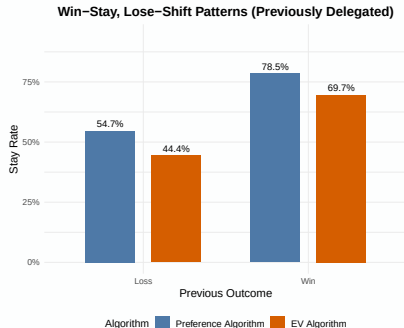
# Dynamics: No clean win-stay/lose-shift



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- ▶ Prefs: prior win  $\rightarrow$  slightly *higher* stay probability ( $\beta = 0.043$ ,  $p = 0.19$ ).
- ▶ EV  $\times$  prior win:  $\beta = -0.102$ ,  $p = 0.041 \Rightarrow$  in EV, participants *shift* more after non-zero payoffs than after zeros.
- ▶ Stay rates after losses: 69% (Prefs) vs. 65% (EV).
- ▷ Aggregate dynamics do *not* fit a classic WSLS story; reinforcement appears asymmetric across rules.





- ▶ Online experiment with stylised lotteries  $\Rightarrow$  external validity to field deployments needs care.
- ▶ Mechanism is *not* causally identified: reason items are ex-post and tied to a hypothetical follow-up question.
- ▶ Short-run repeated interaction only; longer-run adaptation to personalised systems is an open question.
- ▶ Categorical non-delegation is an *exploratory* outcome (preregistered analyses are the Boschloo and WMW tests + the LPM).

1. **Preregistered behavioural** comparison of *objective* vs *personalised* decision rules  $\Rightarrow$  personalisation does *not* increase delegation.
2. The effect is concentrated at **first contact** (Round 1: 66.9% EV vs 55.8% Prefs) — and first contact often decides whether users return at all.
3. **Categorical non-delegation** is a distinct uptake barrier under personalisation (15.7% vs 3.4%), *not* explained by demographics, risk attitudes, AI exposure, or dissatisfaction with the elicited profile.

*The very feature meant to increase subjective fit  
may encroach on users' sense of choice authority,  
deterring precisely those who value exercising active control.*

Thank you!

Thank you — and looking forward to your  
questions.

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Preregistration: <https://aspredicted.org/7un5qf.pdf>  
Paper, data & code available upon publication.